

ZQUAS POSITION PAPER · APRIL 2026

# Beyond Banking: Cross-Sector Federated Detection for Financial Crime, Telecommunications Fraud, and Digital Asset Compliance

A single detection engine that correlates risk signals across banks, telecommunications operators, and digital asset platforms without sharing raw data between institutions. The same code, the same cryptographic protocols, the same GPU-native detection pipeline, applied to every regulated sector where financial crime operates.

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## 1. The Cross-Sector Problem

Financial crime does not respect institutional boundaries. It does not respect sectoral boundaries either.

An Authorised Push Payment fraud starts with a phone call. The fraudster calls the victim, impersonating their bank. The call is routed through a telecommunications network. The victim transfers funds to a "safe account" controlled by the fraudster. The money moves through two or three bank accounts within minutes. Within hours, it exits through a cryptocurrency exchange into a stablecoin, crosses a blockchain bridge, and arrives in a jurisdiction with no bilateral cooperation agreement.

The telecommunications operator saw the call. It came from a number that contacted 47 different people in the past week. The telco did not share this information with the bank.

The bank saw the payment. A high-value transfer to a first-time payee, initiated within two hours of an inbound call. The bank did not know about the call because it has no access to telecommunications data.

The cryptocurrency exchange saw the deposit. Funds arriving from a bank account opened three weeks ago, immediately converted to stablecoin and bridged to another chain. The exchange did not know the funds originated from an impersonation fraud.

**Three institutions. Three data domains. Three independent compliance systems. Each one sees a fragment. None of them sees the crime.**

According to the UK Finance Annual Fraud Report 2025, criminals stole GBP 1.17 billion through payment fraud in 2024, of which GBP 450.7 million was lost to APP fraud alone. 70 per cent of APP fraud cases started online. 16 per cent originated through telecommunications networks.

**The criminals have cross-sector visibility. The compliance infrastructure does not.**

## 2. What Exists Today

The ZQUAS F1 Engine is a GPU-native financial crime detection engine of approximately 753,000 source lines excluding third-party code. Readiness is TRL 6: an integrated prototype demonstrated in a controlled relevant environment,

validated on synthetic populations, with no production deployments.

2.1 Banking Detection: Built and Tested

The engine processes 500,000 entities in under 2 seconds with an alert lifecycle under 10 milliseconds. Total VRAM consumption is under 1 gigabyte. It runs on hardware ranging from a consumer-grade laptop GPU to data centre GPUs (A100, H100). The detection pipeline evaluates five layers covering temporal analysis, structural graph analysis, behavioural profiling, corridor analysis, and flow path reconstruction. All five layers execute on GPU with over 100 compiled compliance policies.

2.2 Bank-to-Bank Federation: Production-Ready

The federation protocol implements Elliptic Curve Diffie-Hellman Private Set Intersection (ECDH-PSI) for entity matching, bilateral escalation signal propagation, and cryptographic evidence attestation. Federation rounds complete in under 10 seconds per bilateral pair.

2.3 Financial Crime Network Simulator: Production-Ready

The F1 Engine includes a GPU-native simulation environment that generates synthetic banking populations with embedded criminal networks. 15 criminal typologies (8 AML, 7 APP fraud) generate realistic transaction data on GPU. The simulator is calibrated against published national statistics for three jurisdictions (Netherlands, United Kingdom, Germany). Over 300 tests pass.

2.4 Maturity Summary

| CAPABILITY                              | STATUS                                 |
|-----------------------------------------|----------------------------------------|
| Banking transaction ingestion           | Built and tested (synthetic data).     |
| Bank-to-bank federation (ECDH-PSI)      | Built and tested (synthetic data).     |
| Five-layer detection pipeline (GPU)     | Built and tested. 56-policy catalogue. |
| FCNS (15 typologies)                    | Built and tested (synthetic data).     |
| Evidence system + blind assessment      | Built and tested (synthetic data).     |
| Deterministic entity hashing            | Built and tested (synthetic data).     |
| Cryptographic erasure (per-entity keys) | Built and tested (synthetic data).     |
| Differential privacy on federation      | Architecture designed. Phase 2.        |
| Signal velocity circuit breaker         | Architecture designed. Phase 2.        |
| Cross-sector score calibration          | Architecture designed. Phase 2.        |
| Telco ingestion (CDR/TMDR)              | Planned. Phase 2 (Q3 2026).            |
| Blockchain ingestion (EVM/UTXO)         | Planned. Phase 3 (Q4 2026).            |

Everything in this paper that is described as "architecture" or "planned" is clearly labelled as such. Everything described without qualification is built, tested, and operational.

3. The Regulatory Direction

Regulators are converging on cross-sector responsibility. The evidence is unambiguous.

3.1 Singapore: Shared Responsibility Framework

Singapore's MAS and IMDA implemented the Shared Responsibility Framework on 16 December 2024. It is the first regulatory framework to hold both financial institutions and telecommunications operators jointly liable for fraud losses. The framework operates on a waterfall principle. MAS added a fraud surveillance duty effective 16 June 2025, requiring banks to detect and block suspicious transactions in real time.

3.2 United Kingdom: Converging Frameworks

The PSR's APP fraud reimbursement regime (7 October 2024) requires PSPs to reimburse victims up to GBP 85,000 within five business days. The UK Telecommunications Fraud Sector Charter (5 November 2025) commits telcos to cross-industry intelligence sharing. Ofcom's consultation on combatting mobile messaging scams (29 October 2025) proposes mandatory KYC for business message senders. The Data (Use and Access) Act 2025 establishes crime and fraud prevention as a lawful basis for data sharing. The UK Fraud Strategy 2026-2029 calls for cross-sector collaboration, supported by the new Operational Crime Centre launching April 2026.

3.3 European Union: AMLR Article 75 and MiCA

AMLR Article 75 mandates mechanisms for cross-institutional information sharing. AMLA assumes oversight from 2027. MiCA became fully applicable in December 2024. CASPs now have the same compliance obligations as banks. When a stablecoin payment originates at a bank, transits a blockchain, and settles at an exchange, every entity in the chain is regulated. The technology to correlate signals across those steps does not exist.

3.4 SWIFT: The Institutional Blockchain Transition

SWIFT announced on 29 September 2025 that it will add a blockchain-based shared ledger to its infrastructure, developed with over 30 financial institutions. When institutional settlement moves onto permissioned chains, the public transparency that Chainalysis and Elliptic rely on disappears. Consensus is not detection. Detection requires behavioural analysis across the entity graph over time. The cross-sector gap remains even when the intra-sector gap narrows.

4. The Architecture

4.1 Design Principle

The F1 Engine operates on entities and relationships. A bank customer who is also a mobile subscriber who is also a crypto exchange user is one entity with three institutional views. At the federation layer, the engine treats all data sources uniformly: a sender, a receiver, a value, and a timestamp.

4.2 The Four-Installation Federation

|                                                          |
|----------------------------------------------------------|
| <b>BANKING INSTITUTION</b>                               |
| F1 Edge Installation in bank's data centre.              |
| Transaction monitoring, entity risk assessment.          |
| Status: Built and tested on synthetic data. TRL 6.       |
| <b>TELECOMMUNICATIONS OPERATOR</b>                       |
| F1 Edge Installation in telco's data centre.             |
| CDR/TMDR analysis, scam pattern detection.               |
| Status: Planned, Phase 2 (Q3 2026).                      |
| <b>DIGITAL ASSET PLATFORM</b>                            |
| F1 Edge Installation at exchange.                        |
| On-chain/off-chain analysis, bridge and mixer detection. |
| Status: Planned, Phase 3 (Q4 2026).                      |

### REGULATORY OBSERVER

Read-only interface for regulators.

Aggregated metrics, detection rates, false positive rates.

No entity-level data. Measures quality, not just quantity.

## 4.3 Cross-Sector Entity Matching

The critical technical challenge is linking the same person across institutional boundaries without sharing personal data. In the target launch jurisdictions, a single government identifier serves as the natural join key. In the Netherlands, the BSN is mandatory at banks (Wwft), telcos (SIM registration), and crypto exchanges (Wwft). The UK uses the National Insurance number and passport.

The F1 Engine uses ECDH-PSI to discover common entities without revealing identity data. The protocol is GPU-accelerated and provably secure under standard cryptographic assumptions.

### Two-Tier Entity Matching:

- **Tier 1:** Government ID hash. Primary mechanism for AML detection. Same person laundering across institutions using the same identity.
- **Tier 2:** Phone number hash (E.164). Primary mechanism for APP fraud. Links the victim across telco and bank. The fraudster and mule are different people. The victim is the join key.

## 4.4 The Normalisation Pipeline

PSI operates on exact hash matches. Deterministic normalisation before hashing (case, punctuation, whitespace, abbreviation, ISO formatting) ensures consistent matching. The pipeline runs locally at each institution before any data enters the cryptographic protocol.

## 4.5 Signal Propagation and Privacy Protection

Each installation computes local risk assessments. When an entity warrants cross-institutional attention, an escalation signal propagates to matched peers. No raw transactions, CDRs, blockchain addresses, or customer names cross institutional boundaries.

*In the current production architecture, federation signals are binary escalations: escalated or not escalated. Binary signals are mathematically immune to threshold drift. There is no intermediate value that noise, rounding, or miscalibration can degrade.*

Escalation signals are batched into federation rounds that aggregate multiple events. The receiving institution cannot determine which specific event triggered the escalation or when it occurred. This defeats inference attacks based on timing correlations.

## 4.6 Intersection Privacy and Anti-Leakage Controls

Three controls mitigate commercial sensitivity of intersection metadata: (1) Risk-filtered PSI: only entities exceeding a local risk threshold enter the PSI set. (2) Ephemeral intersection: matched identifiers are discarded after the federation round. (3) PSI query budgets: limits on rounds per bilateral pair per period prevent incremental customer-base mapping.

## 4.7 Cross-Sector Score Calibration (Planned)

The simulator generates synthetic populations with cross-sector activity. Calibration is computed bilaterally. Both institutions verify the mapping against the same ground truth. Bilateral calibration floors ensure poorly calibrated installations improve before influencing the federation. The signal velocity circuit breaker catches aggressive installations. The calibration floor catches negligent ones. Together they bound signal quality from both directions.

#### 4.8 The APP Fraud Detection Example: Victim-Centric Linkage

In professional APP fraud, the fraudster and the mule are different people. The architecture centres linkage on the victim.

##### Step 1: Telco detects scam pattern.

Burner SIM calls 200 potential victims. Burst calling pattern detected. Each targeted subscriber flagged locally. No data leaves the telco.

##### Step 2: Federation round: PSI match on phone number.

Victim's phone number appears in both the telco's risk set and the bank's customer set. Match identifies the victim, not the fraudster.

##### Step 3: Escalation signal propagates to bank.

"This customer is being targeted by a telecommunications scam pattern." No call details, no timing, no number shared.

##### Step 4: Victim initiates payment to mule account.

Bank's detection pipeline combines cross-sector signal + first-time payee + high-value transfer. Alert in under 10ms.

##### Step 5: Payment held. Fraud lifecycle broken.

Only the specific high-risk payment is held. Salary, direct debits, routine transactions continue uninterrupted.

*The architecture does not need to unmask the fraudster to protect the victim. By correlating the targeting signal at the telco with the transfer signal at the bank, the fraud lifecycle is broken before the first payment settles.*

#### 4.9 The Private Blockchain Detection Example

A bank customer initiates a GBP payment through SWIFT's blockchain-based ledger. The recipient converts the stablecoin to a different token via a regulated bridge. The tokens are sent to a mixer. On the permissioned chain: only participating banks see the transaction. On the public chain: no link back to the originating bank payment.

With cross-sector federation: the bank's F1 installation assesses the originating entity. The CASP's F1 installation assesses the bridge user. PSI matches the entity across both (same government ID at KYC). The CASP's escalation signal propagates to the bank. Without federation: neither connects the dots.

#### 4.10 Evidence Attestation

The regulatory observer receives aggregated metrics. When a case requires law enforcement action, the F1 Engine generates an evidence package: investigation items from the detecting institution's own data, a cryptographic attestation transcript signed by each participating installation, and sector attribution indicating which sectors contributed signals. The prosecutor can demonstrate cross-sector criminal activity without the institutions ever having seen each other's files.

#### 4.11 Protocol Governance

Nobody decides what to detect. Everybody decides how to share. Each institution compiles its own detection policies. They only agree on how to share the output: entity-level escalation signals. Protocol governance is limited to: data format versioning, federation round timing, PSI parameters, privacy budgets, query limits, and signal quality floors.

### 5. GDPR Compliance Architecture

- **No raw data sharing:** No customer data, transaction records, CDRs, or blockchain addresses cross boundaries. Only binary escalation signals.

- **Cryptographic matching:** ECDH-PSI blinds all identity data before comparison. Provably secure under standard assumptions.
- **Lawful basis:** Data (Use and Access) Act 2025 and AMLR Article 75 provide the foundation for crime prevention data sharing.
- **Cryptographic erasure:** GDPR Article 17 right to erasure via hierarchical per-entity key deletion. Signed destruction receipts.
- **Purpose limitation:** Escalation signals do not reveal the underlying reason. Bank learns "elevated risk from telecommunications" but not the specific activity.
- **No mass surveillance:** PSI operates on risk-filtered entity sets, not full customer bases. No central database. Each institution retains full sovereignty.

## 6. The Telco Value Proposition

Singapore has implemented shared liability. The UK Fraud Sector Charter commits telcos to cross-industry intelligence sharing. Australia's Scam Prevention Framework imposes penalties up to AUD 50 million for non-compliance. The direction is global and irreversible.

Under shared liability frameworks, not participating is more damaging than participating. A telco that could have flagged risk but did not faces financial liability and regulatory enforcement. The federation protocol shares an escalation signal, not an admission. The telco is demonstrating vigilance, not confessing failure.

According to the CFCA's 2023 Global Fraud Loss Survey, telecommunications fraud losses reached USD 38.95 billion globally in 2023, representing 2.5 per cent of industry revenues. The F1 Edge Installation runs inside the telco's data centre, on the telco's hardware, under the telco's operational control.

## 7. The Crypto Exchange Value Proposition

MiCA requires CASPs to implement transaction monitoring equivalent to traditional financial institutions. Most exchanges check wallet blacklists. They cannot detect novel laundering patterns or the intersection of on-chain and off-chain activity.

Exchanges as members: we are testing a withdrawal check that asks the funding bank one question about the payment that funded it. Results will be published when the test is complete.

The FATF Travel Rule requires originator and beneficiary information to be transmitted with virtual asset transfers. The federation protocol can facilitate compliance by confirming entity identity matches without revealing underlying data.

## 8. The Market Gap

**No company in the world offers a cross-sector federated detection platform.**

Telecommunications fraud vendors (Subex, SEON, Vonage) protect telcos from fraud on their own networks. They do not share detection signals with banks.

Banking AML vendors (NICE Actimize, Feedzai, Tookitaki, Oracle Financial Crime) monitor transactions within a single bank. Some offer consortium models but none include telecommunications or blockchain data.

Blockchain analytics firms (Chainalysis, Elliptic, Merkle Science) analyse public chain data. Their model depends on public blockchain transparency, which will diminish.

ZQUAS is architected to process banking, telecommunications and blockchain data in a single GPU-native detection pipeline, connected by a privacy-preserving federation protocol across all three sectors. In the competitive landscape reviewed, no current system has been identified that performs cross-institutional risk determination without disclosure across those sectors.

## 9. Honest Limitations

- **Federation speed:** Local detection under 10ms. Federation enrichment under 10s per bilateral round. A window of seconds to minutes exists for newly identified signals, vs. hours/days in batch.
- **Adoption:** Network effect requires a minimum viable consortium. Initial deployment targets regulated groups rather than individual institutions.
- **Jurisdiction-specific legal work:** Lawful basis established in principle. Operational boundaries still being defined per jurisdiction.
- **Cross-sector calibration:** No cross-sector detection calibration methodology exists today. Simulation-based approach requires validation through pilot deployments.
- **Multi-hop attenuation:** Bilateral federation propagates one hop. Hub-spoke topology designed but not yet deployed.
- **ID fragmentation:** In jurisdictions without a universal identifier, multi-ID hashing mitigates but does not eliminate false negatives.

## 10. Timeline

### PHASE 1: Current

Banking federation. Pilot validation.

Status: Built and tested on synthetic data. TRL 6.

### PHASE 2: Q3 2026

Telecommunications integration. CDR/TMDR ingestion.

Cross-sector PSI on phone number and government ID.

Differential privacy, signal velocity circuit breaker.

### PHASE 3: Q4 2026

Digital asset integration. Blockchain event ingestion.

Wallet-to-identity linking via exchange KYC.

### PHASE 4: 2027

Regulatory interface. AMLA Article 75 reporting.

Hub-spoke topology for multi-hop propagation.

**AMLR Article 75 applies: July 10, 2027. Cross-institutional information sharing mandated.**

## 11. Conclusion

The financial crime industry has spent twenty years optimising detection within institutional silos. The criminals have spent twenty years exploiting the gaps between those silos. The gaps are now wider than ever: spanning banking, telecommunications, cryptocurrency, and private institutional blockchains.

The regulatory response is converging. Singapore has implemented cross-sector liability. The UK is building the framework. The EU mandates cross-institutional sharing under AMLR. MiCA brings crypto exchanges into the regulated perimeter.

The technology to match this regulatory direction did not exist. A detection engine that operates across banking, telecommunications, and blockchain data, connected by a federation protocol that shares escalation signals without sharing raw data, running at the speed required for real-time intervention, governed by the institutions that use it rather than a central authority.

**We do not need to unmask the fraudster to protect the victim. By correlating the targeting signal at the telco with the transfer signal at the bank, we break the fraud lifecycle before the first payment settles.**

The banking detection engine is built. The federation protocol is built. The simulation environment is built. The cross-sector extensions are engineering milestones on an operational foundation.

**It exists now.**

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*ZQUAS. The criminals have cross-sector visibility.*

*Now the compliance infrastructure does too.*

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